

Indian Institute of Technology, Kharagpur



Department of Electronics & Electrical Communication Engineering

Subject: Communication I Laboratory

Year: 2026

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Experiment 6 : Design of IIR Butterworth LPF, BPF and FIR filters using Hamming window method.

OBJECTIVE:

1. To design IIR Butterworth Low Pass Filter (LPF) and Band Pass Filter (BPF) in MATLAB.
2. To design FIR filters using the Hamming window method in MATLAB.
3. To analyze magnitude and phase responses of the designed filters.
4. To apply the filters on the audio signal and observe the spectrum before and after filtering.

SPECIFICATIONS:

LPF: Cutoff frequency=4.5kHz
Sampling frequency=20kHz
IIR filter order=3
FIR filter order=20
Passband ripple factor=0.1
Stopband ripple factor=0.5

BPF: Higher cutoff frequency=12kHz
Lower cutoff frequency=8kHz
Sampling frequency=40kHz
IIR filter order=3
FIR filter order=20
Passband ripple factor=0.1
Stopband ripple factor=0.5

PROCEDURE:

Design of LPF and BPF in MATLAB

1. First, calculate the normalized cutoff frequency using the cutoff frequency and the sampling frequency. The normalized frequency is given by

$$W_n = F_c / (F_s / 2)$$

where F_c is the cutoff frequency and F_s is the sampling frequency.

2. Using this normalized frequency, design a 3rd order Butterworth Low Pass IIR filter, Butterworth Band Pass IIR filter in MATLAB to obtain the filter coefficients.
3. Next, design the FIR filter using the truncation(Hamming window method).

$$w(n) = 0.54 - 0.46 \cos(2\pi n / (N-1))$$

The maximum FIR filter order is limited to 20, and the filter order should be optimized so that the ripple in the passband does not exceed the limits specified in the filter specifications.

4. Plot the magnitude response (Magnitude in dB vs Frequency in kHz) and the phase response (Phase in radians vs Frequency in kHz) of the designed filters to analyze their frequency characteristics.
5. Load the audio signal  audio.mp3 and import it into MATLAB using the audioread command. The audio signal is then resampled to the target sampling frequency(i.e.LPF =20kHz, BPF=40kHz).
6. Apply the designed IIR and FIR filters to the audio signal.
7. Plot the magnitude and phase responses of the signal before filtering and after filtering to observe the effect of the filter on the signal spectrum.
8. Store the filtered audio signal for further analysis and comparison.
9. Finally, verify and compare the unfiltered and filtered signals using the online spectrum analyzer tool available at: <https://academo.org/demos/spectrum-analyzer/>

OUTPUTS:

IIR Low pass filter coefficients:

Denominator Coefficients:

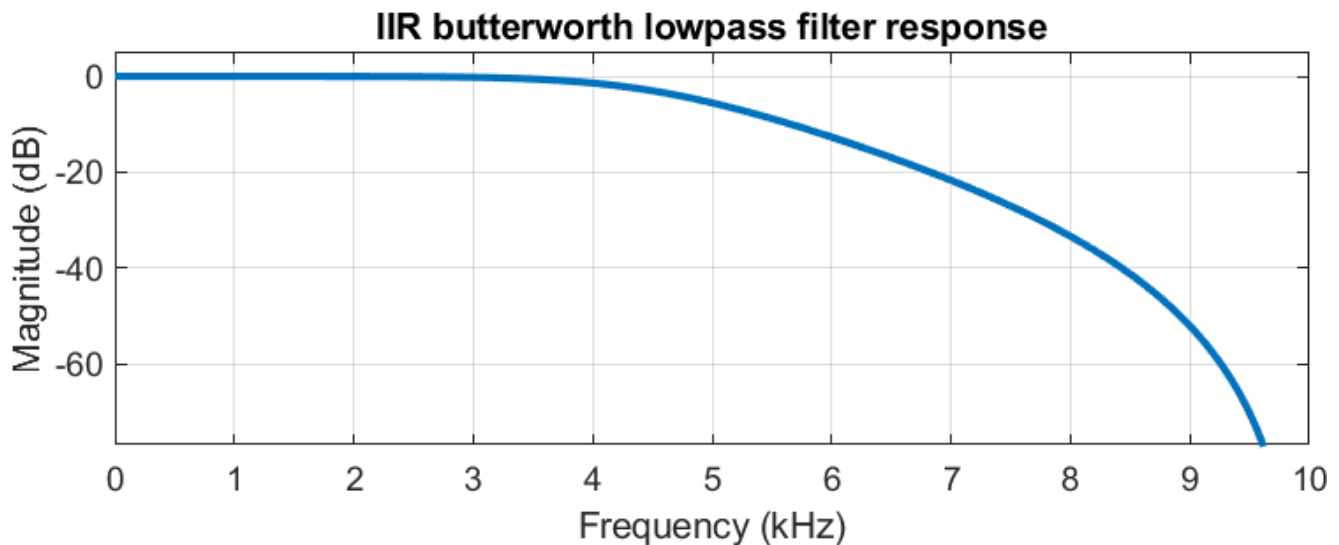
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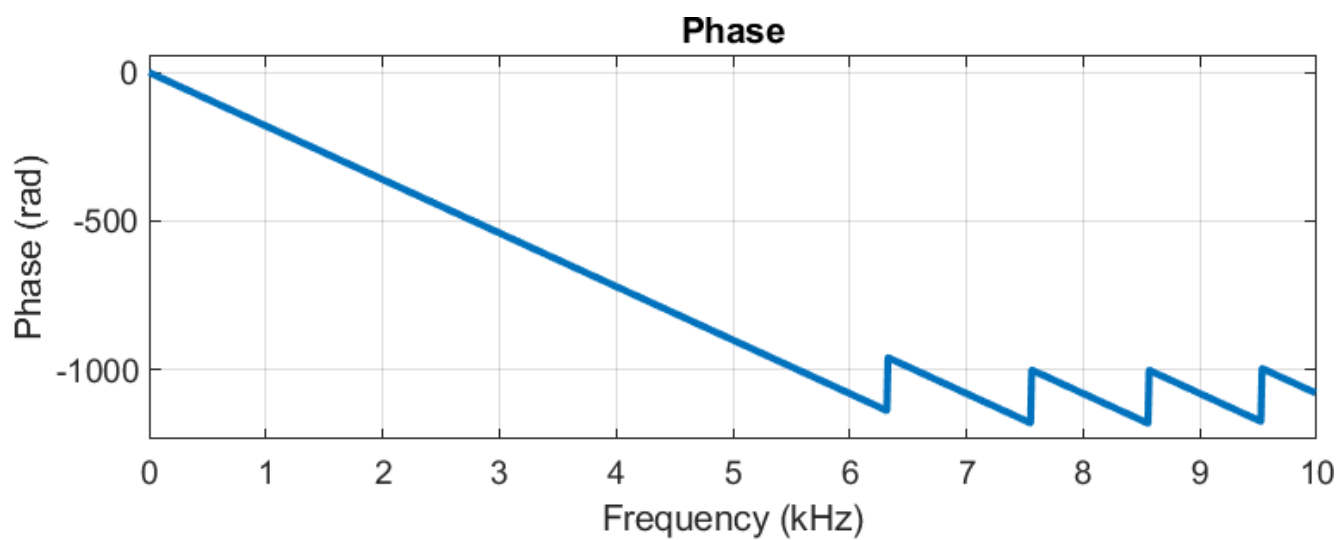
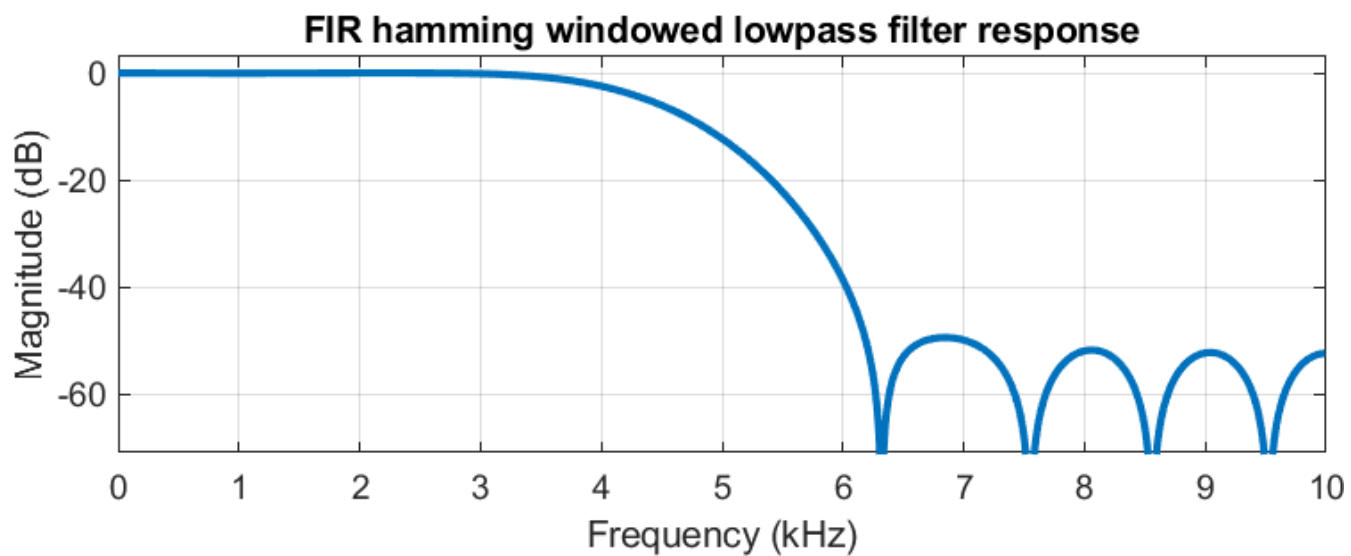
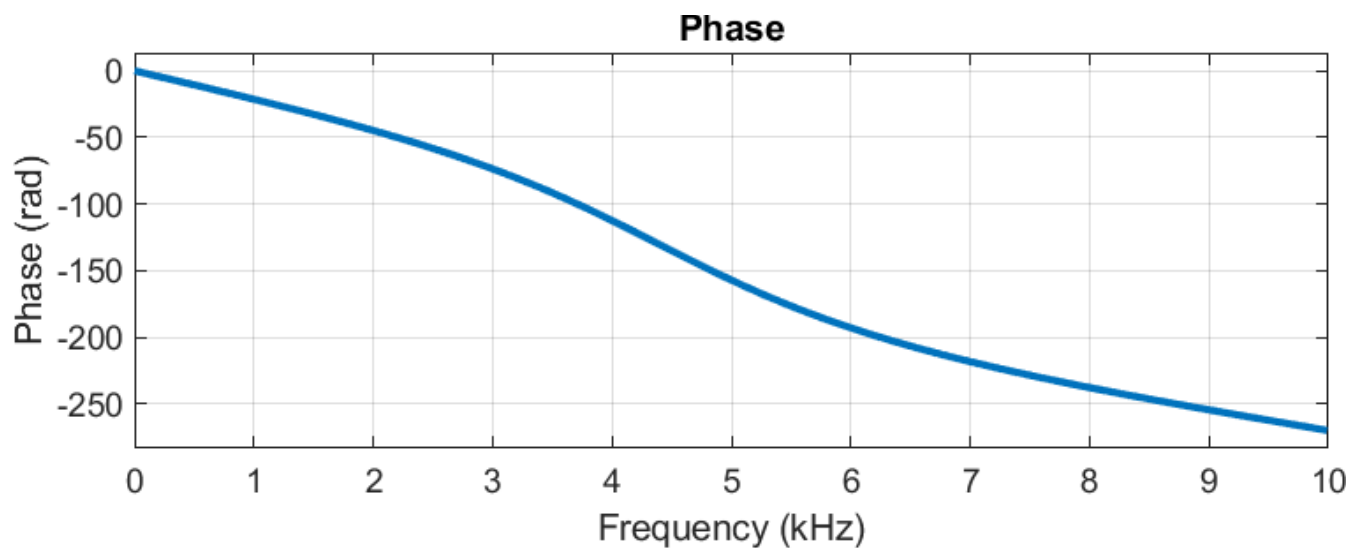
Numerator Coefficients:

b_iir:[0.130063011757311, 0.390189035271933, 0.390189035271933,
0.130063011757311]

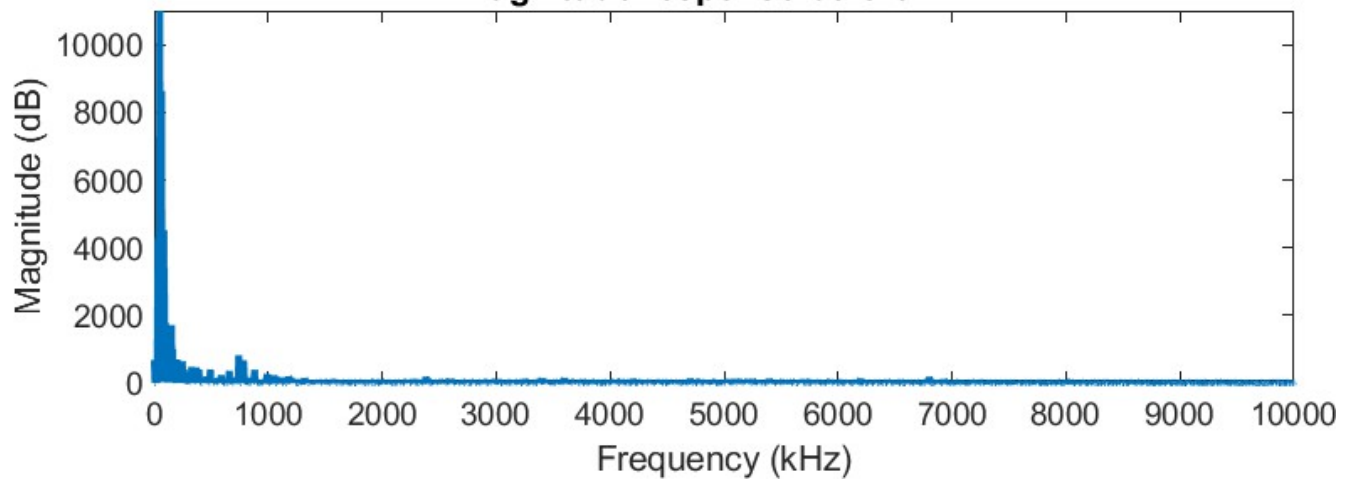
FIR Low pass filter coefficients (Numerator Coefficients):

b_fir:[0.00254020885588450, 0.000565786474192167, -0.00633611090078127,
-0.00555236885155625, 0.0170336444532568, 0.0242486852529517,
-0.0318285333307475, -0.0764237658295615, 0.0447504112200591,
0.306556063524820, 0.448891958262965, 0.306556063524820,
0.0447504112200591, -0.0764237658295615, -0.0318285333307475,
0.0242486852529517, 0.0170336444532568, -0.00555236885155625,
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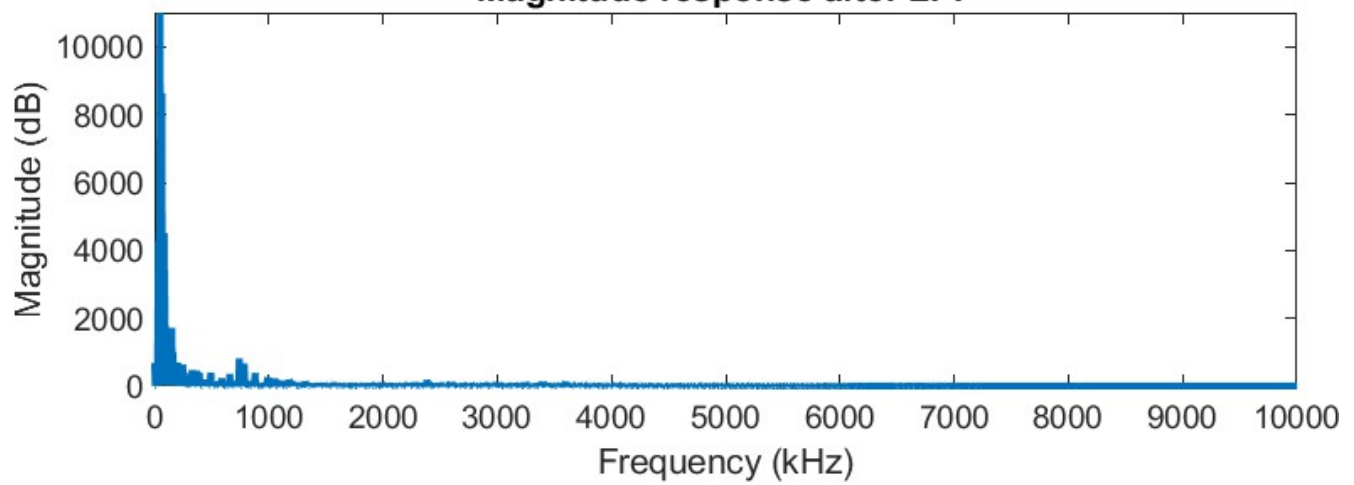




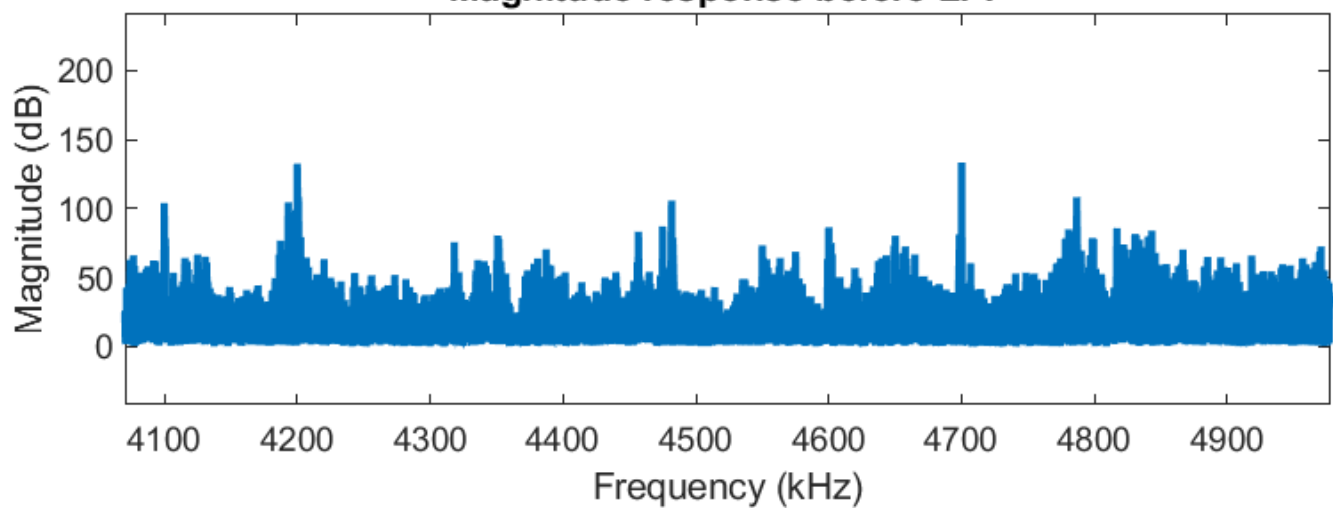
Magnitude response before LPF

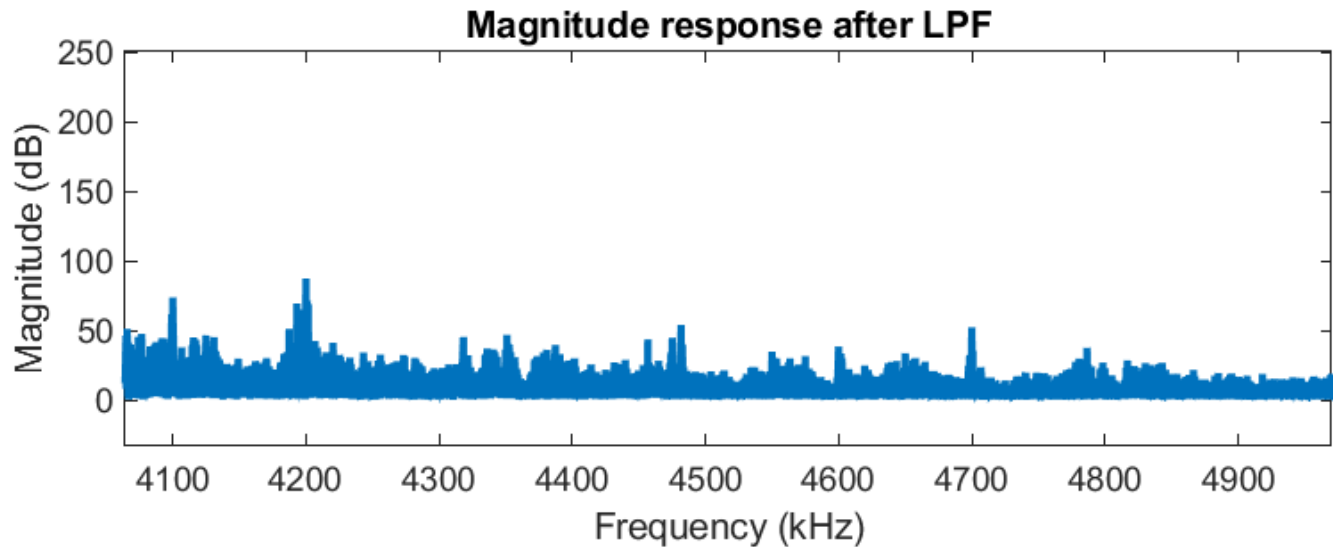


Magnitude response after LPF



Magnitude response before LPF





IIR Band pass filter coefficients:

Denominator Coefficients:

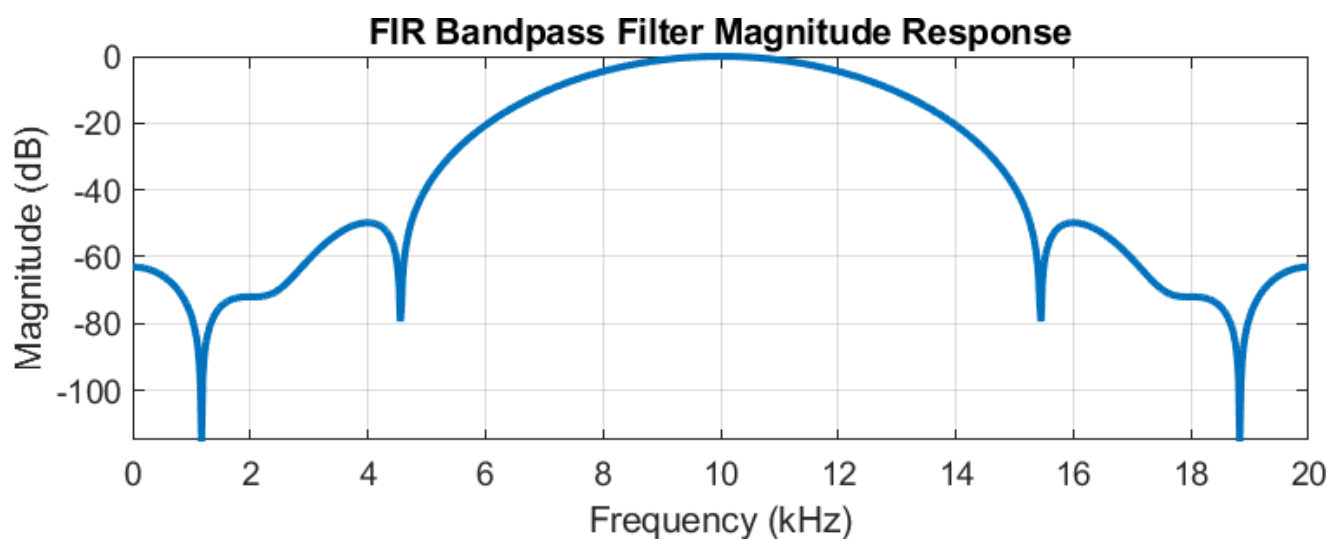
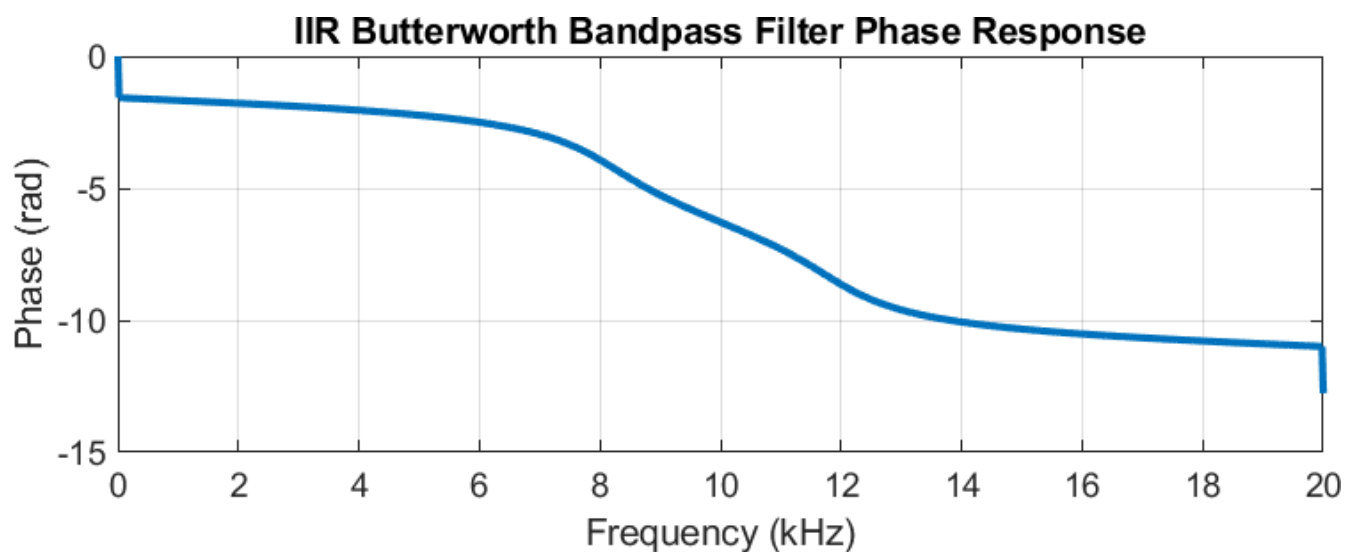
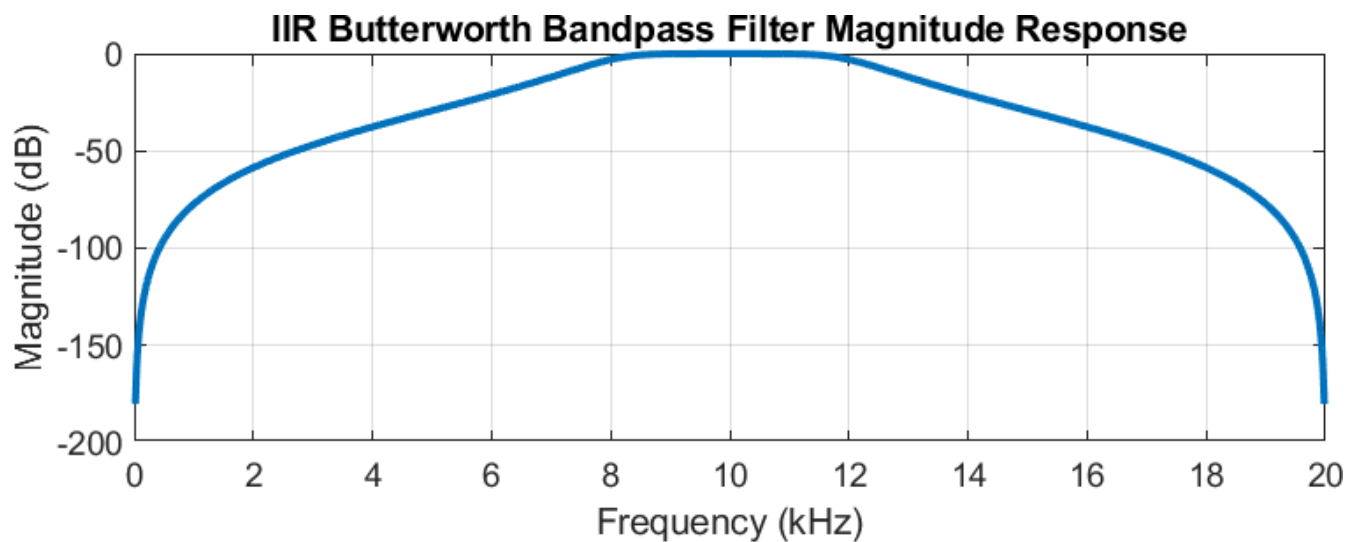
a_iir: [1, -2.05391259555654e-15, 1.76004188034317, -2.85882428840978e-15,
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Numerator Coefficients:

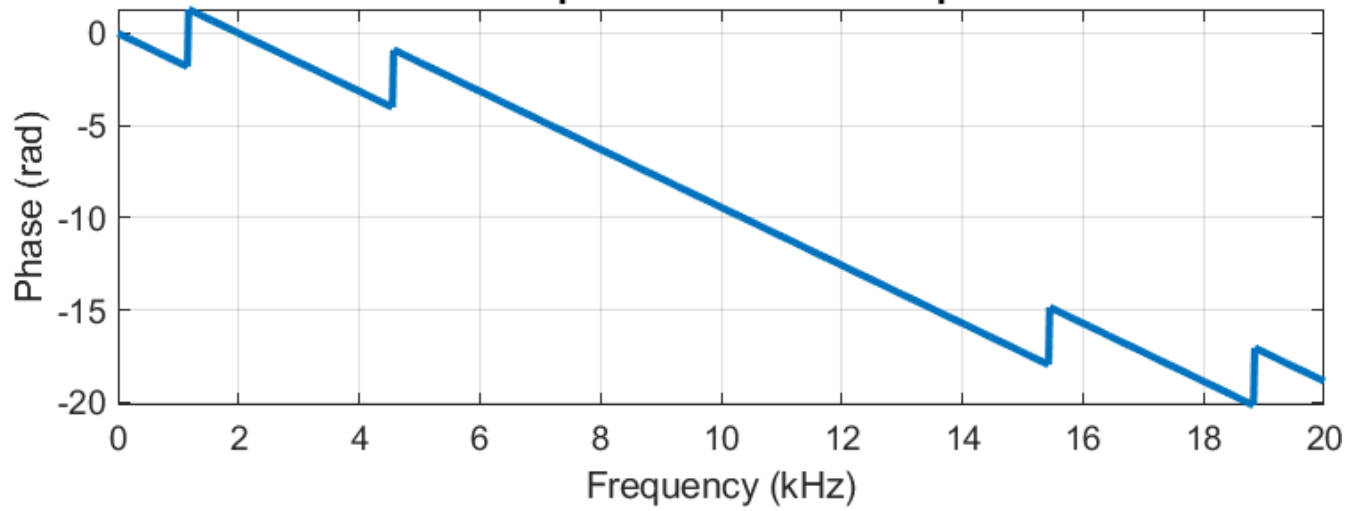
b_iir: [0.0180989330075144, 0, -0.0542967990225433, 0,
0.0542967990225433, 0, -0.0180989330075144]

FIR Band pass filter coefficients(Numerator Coefficients):

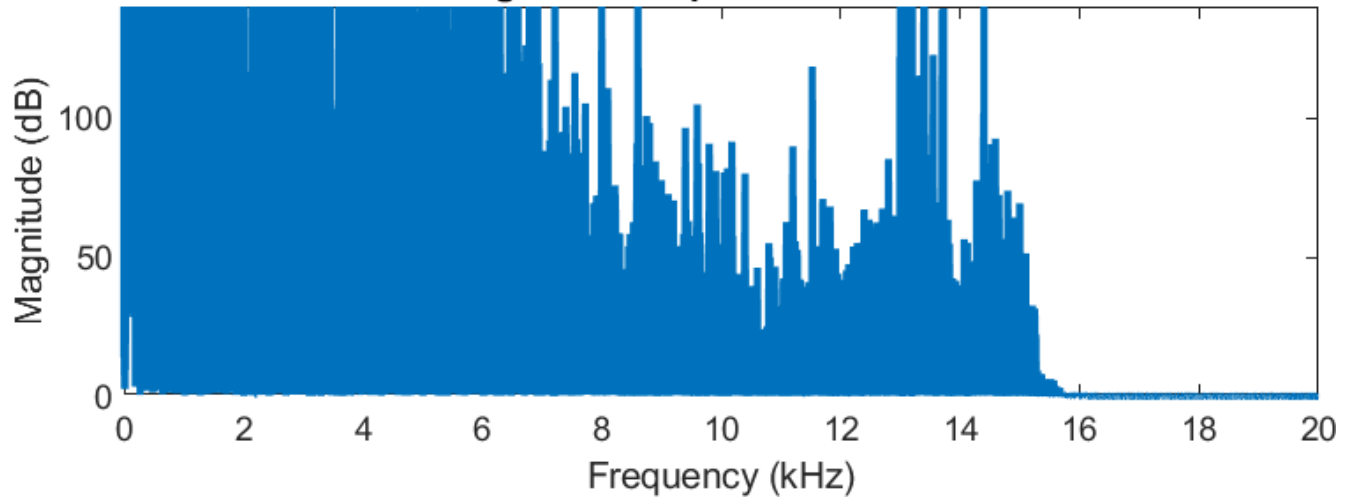
b_fir: [-7.39142315096372e-19, 1.68597269678491e-18, 0.00930428314501814,
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,0.122363546361145, -7.99664770204792e-17, -0.202246558429840,
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2.49460531345026e-17, -0.0475777661344175, -9.97700088405679e-18,
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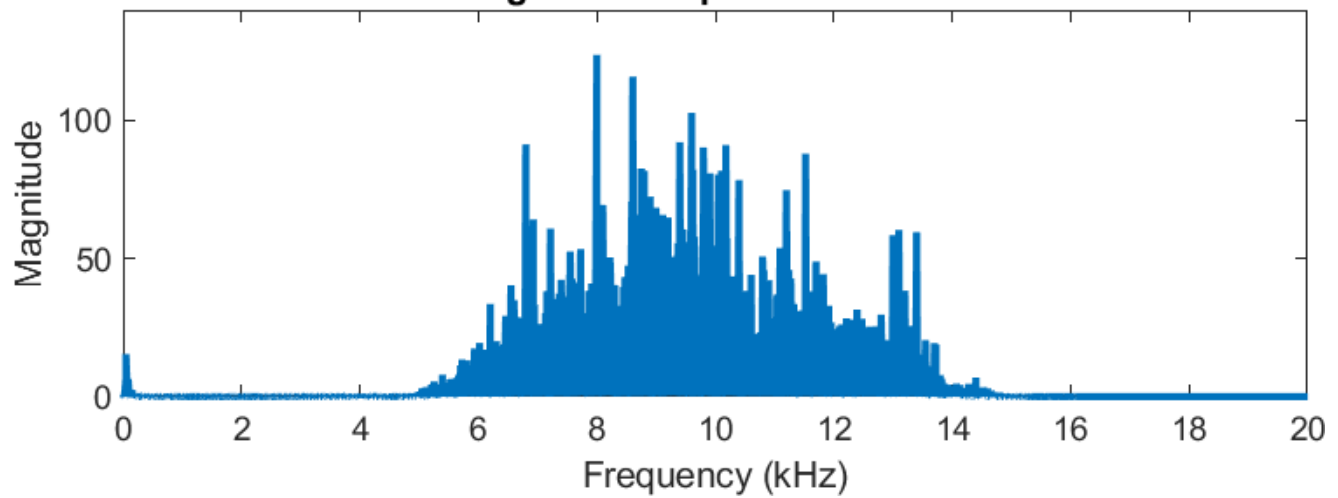
FIR Bandpass Filter Phase Response



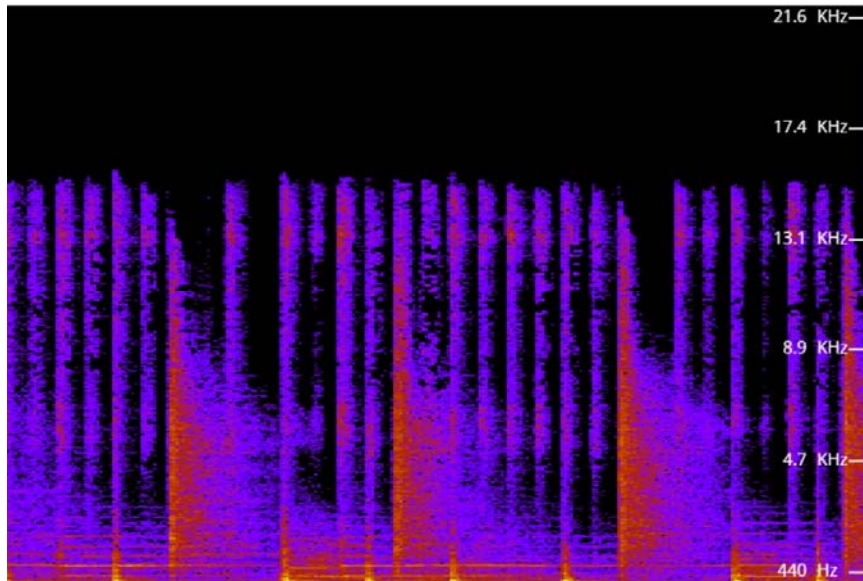
Magnitude response before BPF



Magnitude response after BPF



Audio signal spectrum study before and after filtering:

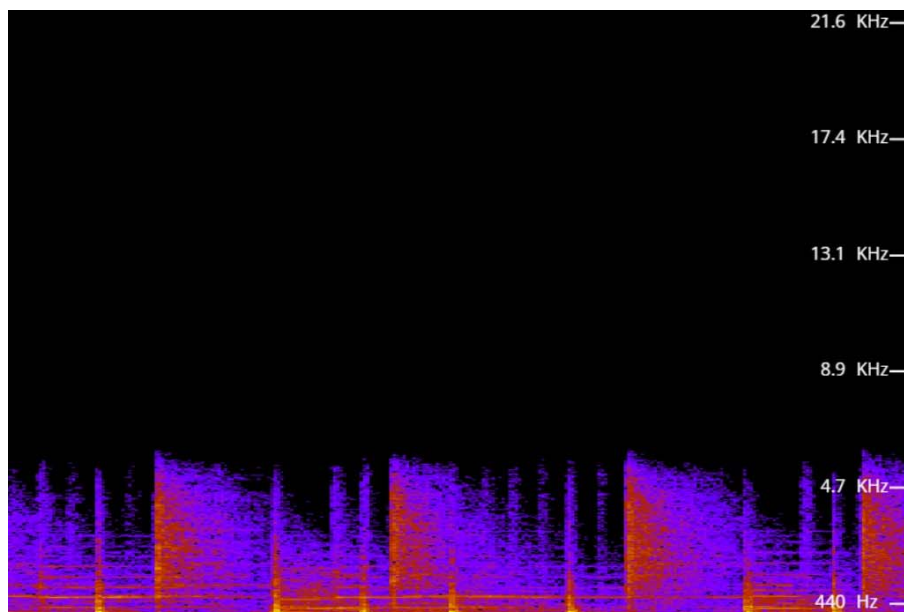


Logarithmic Frequency Scale?
☐

Sound Sample
Bird Song (Song Thrush) ▾

Upload your own
Choose File audio.mp3

▶ 0:04 / 0:24

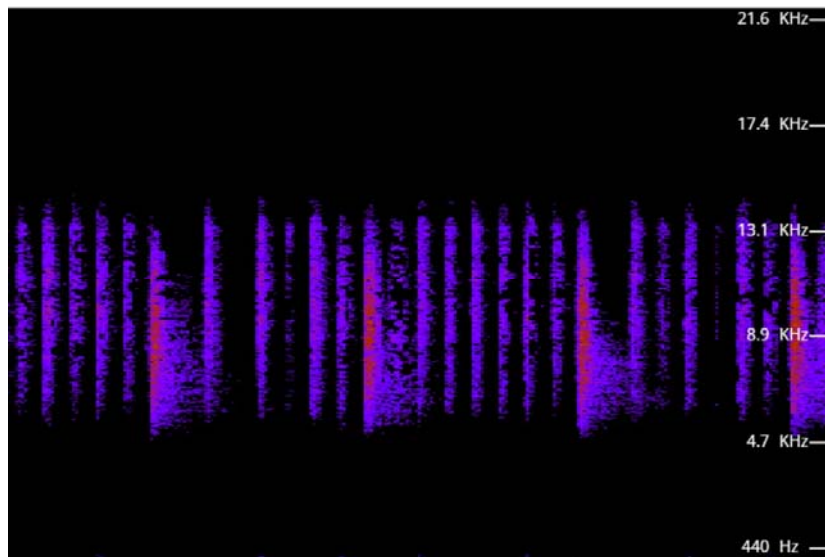


Logarithmic Frequency Scale?
☐

Sound Sample
Bird Song (Song Thrush) ▾

Upload your own
Choose File LPF filtered aud

▶ 0:05 / 0:24



Logarithmic Frequency Scale?
☐

Sound Sample
Bird Song (Song Thrush) ▾

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▶ 0:05 / 0:24